

A Perfect Secret Sharing Scheme for r -Uniform Hypergraph-Based Access Structures

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Abstract

A secret sharing scheme is a method to distribute a secret among a set of participants, such that only qualified subsets of the participants can recover the secret. A secret sharing scheme is perfect if any unqualified subset obtains no information regarding the master key. The collection of qualified subsets is called access structure. In a hypergraph, if the size of edges is equal to r , the hypergraph is called an r -uniform hypergraph. An r -uniform hypergraph-based access structure is an access structure which using an r -uniform hypergraph to show the access structure, where a vertex denote a participant and the edge set denote the minimal access structure of a secret sharing scheme. Given any r -uniform hypergraph G and Γ is the G -based access structure, this paper proposes a perfect secret sharing scheme for Γ , and this scheme is the most efficient secret sharing scheme for Γ up to now.

1 Introduction

In 1979, Blakley [1] and Shamir [13] independently introduced the concept of secret sharing. A secret sharing scheme is a method to distribute a secret among a set of participants P in such a method that only certain pre-specified subsets, called *qualified* (or *authorized*) subsets, of the participants can reconstruct the secret and unqualified subsets can not. Another view of a secret sharing scheme is a pair of efficient algorithms: *distribution algorithm* and *reconstruction algorithm* run by a dealer and some participants. The distribution algorithm is executed by a dealer who, given a secret, broken it into pieces, called *shares*, and gives them to the participants. The reconstruction algorithm is executed by a qualified subset of participants who, by putting together their own shares, can reconstruct the secret. The collection of qualified subsets is called *access structure* (denoted by Γ); the collection of unqualified subsets is called *prohibited structure* (denoted by Δ).

If any set in the prohibited structure not only can not recover the secret, but also can not know as much information regarding the secret as they have no any shares, then the scheme is called *perfect* [6, 8, 10, 13]. Let $P = \{p_1, p_2, \dots, p_n\}$ be the set of all participants and 2^P be the set of all subsets of P . Let $\mathcal{K}$ be the secret space, $\mathcal{S}$ be the share space for all participant and $\mathcal{S}_i$ be the set of share space that p_i might receive. Define $\rho_i = \log|\mathcal{K}| / \log|\mathcal{S}_i|$, for $1 \leq i \leq n$. The efficiency of a secret sharing scheme is measured by the information rate ρ [3], defined as $\rho = \min \{\rho_i : 1 \leq i \leq n\} = \log|\mathcal{K}| / \log|\mathcal{S}|$. In the implementation of secret sharing schemes, an important issue is to minimize the number of shares distributed to each participant. In other words, we want to make the information rate as large as possible.

For any qualified subset $A \in \Gamma$, any superset of A is also an qualified subset intuitively. Hence, the access structure should satisfy the *monotone increasing* property, and the prohibited structure satisfies the *monotone decreasing* property:

$$A \in \Gamma \text{ and } A \subseteq B \subseteq P \Rightarrow B \in \Gamma.$$

$$A \in \Delta \text{ and } B \subseteq A \subseteq P \Rightarrow B \in \Delta.$$

Let Γ_0 be a family of the minimal sets in Γ , called the *minimal access structure* which is defined by $\Gamma_0 = \{A \in \Gamma : A' \not\subset A \text{ for all } A' \in \Gamma - \{A\}\}$. The family of maximal sets in Δ , called *maximal prohibited structure*, is denoted by Δ_1 . That is, $\Delta_1 = \{B \in \Delta : B \not\subset B' \text{ for all } B' \in \Delta - \{B\}\}$.

In [19], Sun and Shieh classified secret sharing scheme into the following three types:

Type I: access structure Γ

For every secret sharing scheme with an access structure Γ in type I, only subsets of participants in Γ can recover the secret key K , others can not. So that prohibited structure $\Delta = 2^P \setminus \Gamma$ is implied.

Type II: prohibited structure Δ

For every secret sharing scheme with a prohibited structure Δ in type II, only subsets of participants in Δ can not recover the secret key K , others can. So that access structure $\Gamma = 2^P \setminus \Delta$ is implied.

Type III: mixed structure (Γ, Δ)

For every secret sharing scheme with a mixed

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structure (Γ, Δ) in type III, subsets of participants in Γ can recover the secret key K and in Δ can not. So that $\Gamma \cap \Delta = \emptyset$ is implied. The other subsets which are not in Γ and Δ may either can recover the secret key or not. That is, $2^P \setminus (\Gamma \cup \Delta)$ can be empty or not.

A *hypergraph* is an ordered pair of disjoint sets $G = (V, E)$ such that V is a finite nonempty set and E is a collection of subsets of V . The set V is the *vertex set* of G and the set E is the *edge set* of G . A hypergraph is *simple* if $e \subseteq f$ implies $e = f$ for all $e, f \in E$. All hypergraphs in this work are simple. A hypergraph is *r-uniform* if the size of any edge e in E is equal to r (≥ 2). Note that a graph is a 2-uniform hypergraph. The *degree* of a vertex v in a hypergraph $G = (V, E)$ is defined as $\deg(v) = |\{e \in E : v \in e\}|$, and the *maximum degree* of G is defined as $d(G) = \max\{\deg(v) : \text{for all } v \in V\}$.

There are many papers discussed the problem about secret sharing schemes. In some of that, the authors gave some secret sharing schemes for graph-based access or prohibited structures. Given a graph, every vertex can be viewed as a participant; every edge is a relation of two participants by that the secret key K can be recovered or not [3, 18, 15]. Generally, let P be the set of participants, and G be a hypergraph with vertex set P and edge set E . In a perfect secret sharing scheme for the access structure based on G , the access structure is denoted by $\Gamma = \{A \subseteq P \mid e \subseteq A \text{ for some } e \in E\}$, and then the prohibited structure is decided by $\Delta = 2^P \setminus \Gamma = \{A \subseteq P \mid e \not\subseteq A \text{ for all } e \in E\}$. In the other words, E is the minimal access structure Γ_0 .

In 2004, Tochikubo modify the scheme proposed by Ito, Saito and Nishizeki [10], and proposed an efficient secret sharing scheme, called *T scheme* [20]. In 2005, Tochikubo, Uyematsu, and Matsumoto [21] propose an efficient scheme based on qualified subsets. We call this scheme *TUM scheme*. Up to now, TUM scheme is the best scheme for general access structure. Instead of two, an *r-uniform* hypergraph-based access or prohibited structures are more general than graph-based access or prohibited structures, respectively. Therefore, in 2005 and 2006, Weng and Juan [22, 23] proposed two perfect secret sharing schemes for *r-uniform* hypergraph-based prohibited structure, called *r-HP1* and *r-HP2*. The information rate of these two schemes is higher than T and TUM schemes, respectively, and *r-HP2* is better than *r-HP1*. In this paper, a new perfect secret sharing scheme for *r-uniform* hypergraph-based access structures is proposed. The scheme used an *r-uniform* hypergraph to present the minimal access structure. And the information rate of this scheme is greater than the information rate of TUM scheme for the same given access structures.

In Section 2, some useful perfect secret sharing schemes are listed which will be used in the following. Section 3 proposes the main result, a perfect secret sharing scheme for *r-uniform* hypergraph-based access structures. An example will be given and the security of this scheme will be proved in this section. At last, the information rate between the proposed scheme and TUM scheme will be compare and the conclusion will be given in Section 4.

2 Preliminaries

This section will review two perfect secret sharing schemes that will be used in the following sections. Shamir's (t, n) -threshold scheme is a famous perfect secret sharing scheme with information rate = 1; TUM scheme is the best perfect secret sharing scheme for general access structure up to now. The reconstruction algorithms of these two schemes are skipped. For details, refer to [13] and [21], respectively.

2.1 Shamir's (t, n) -threshold scheme

The (t, n) -threshold scheme is a scheme to share a secret key K among P , the set of n participants such that every set of at least t participants can recover the secret key K , but other set of less than k participants can not get any information about the secret key. The (t, n) -threshold scheme is perfect. The access structure Γ of (t, n) -threshold scheme is $\Gamma = \{A : |A| \geq t \text{ and } A \subseteq P\}$; the prohibited structure is $\Delta = \{A : |A| < t \text{ and } A \subseteq P\}$. The Shamir's (t, n) -threshold scheme [13] is described as follows:

There is a finite field $\text{GF}(q)$ in where q is a large prime and assumes key space $\mathcal{K}$ and share space $\mathcal{S}$ are over $\text{GF}(q)$.

- Step 1:** A dealer chooses n distinct nonzero elements over the finite field $\text{GF}(q)$ which is denoted by $x_1, x_2, \dots, x_n$ and values x_i are public for all $1 \leq i \leq n$.
- Step 2:** Suppose the dealer wants to share a secret key K , and then the dealer chooses $(t - 1)$ elements $a_1, a_2, \dots, a_{t-1}$ over $\text{GF}(q)$ independently with the uniform distribution.
- Step 3:** Let $f(x) = a_{t-1}x^{t-1} + \dots + a_2x^2 + a_1x + K \pmod{q}$ be a polynomial of degree $(t - 1)$ over the $\text{GF}(q)$. The dealer distributes the share $S_i = f(x_i)$ to participant p_i for all $1 \leq i \leq n$.

2.2 TUM scheme

In 2005, Tochikubo, Uyematsu and Matsumoto [21] propose an efficient secret sharing schemes,

called TUM scheme. Up to now, TUM scheme is the best scheme for general access structure. For $P = \{p_1, p_2, \dots, p_n\}$, $K \in \mathcal{K}$ and access structure Γ , the TUM scheme is described as follows:

Step 1: Let $\Gamma_0' = \{A \in \Gamma_0: |A| \leq l\}$, where $l = \max_{B \in \Delta} |B|$ and represent it as $\Gamma_0' = \{A_1, A_2, \dots, A_d\}$ with $d = |\Gamma_0'|$.

Step 2: Let $P' = \{p \in X: X \in \Gamma_0 \text{ and } |X| > l\}$ and $n' = |P'|$. Compute n' shares $S = \{s_1, s_2, \dots, s_{n'}\}$ for the secret K by using Shamir's $(l+1, n')$ -threshold scheme. Then, one distinct share in S is assigned to each participant in P' .

Step 3: For every $A_i \in \Gamma_0'$, compute $|A_i|$ shares $S_i = \{s_{n'+i,1}, s_{n'+i,2}, \dots, s_{n'+i,|A_i|}\}$ by using Shamir's $(|A_i|, |A_i|)$ -threshold scheme with K as a secret independently for $1 \leq i \leq d$. One distinct share in S_i is assigned to each $p \in A_i$ ($1 \leq i \leq d$).

For easily understanding, an example by using TUM scheme is given as follows. Let $P = \{p_1, p_2, p_3, p_4, p_5, p_6\}$. The access structure $\Gamma = \{\{p_1, p_2, p_5\}, \{p_1, p_3, p_4\}, \{p_1, p_4, p_5\}, \{p_1, p_2, p_3, p_5\}, \{p_1, p_2, p_4, p_5\}, \{p_1, p_2, p_5, p_6\}, \{p_1, p_3, p_4, p_5\}, \{p_1, p_3, p_4, p_6\}, \{p_1, p_4, p_5, p_6\}, \{p_1, p_2, p_3, p_4, p_5\}, \{p_1, p_2, p_3, p_4, p_6\}, \{p_1, p_2, p_3, p_5, p_6\}, \{p_1, p_2, p_4, p_5, p_6\}, \{p_1, p_3, p_4, p_5, p_6\}, \{p_1, p_2, p_3, p_4, p_5, p_6\}\}$. The prohibited structure $\Delta = \{A \mid A \subseteq P \text{ and } |A| < 3\} \cup \{\{p_1, p_2, p_3\}, \{p_1, p_2, p_4\}, \{p_1, p_2, p_6\}, \{p_1, p_3, p_5\}, \{p_1, p_3, p_6\}, \{p_1, p_4, p_6\}, \{p_1, p_5, p_6\}, \{p_2, p_3, p_4\}, \{p_2, p_3, p_5\}, \{p_2, p_3, p_6\}, \{p_2, p_4, p_5\}, \{p_2, p_4, p_6\}, \{p_2, p_5, p_6\}, \{p_3, p_4, p_5\}, \{p_3, p_4, p_6\}, \{p_3, p_5, p_6\}, \{p_4, p_5, p_6\}, \{p_1, p_2, p_3, p_4\}, \{p_1, p_2, p_3, p_6\}, \{p_1, p_2, p_4, p_6\}, \{p_1, p_3, p_5, p_6\}, \{p_2, p_3, p_4, p_5\}, \{p_2, p_3, p_4, p_6\}, \{p_2, p_3, p_5, p_6\}, \{p_2, p_4, p_5, p_6\}, \{p_3, p_4, p_5, p_6\}\}$. Hence, $\Gamma_0 = \{\{p_1, p_2, p_5\}, \{p_1, p_3, p_4\}, \{p_1, p_4, p_5\}\}$. The TUM scheme is executed as follows.

Step 1: $l = 5$, $\Gamma_0' = \{A_1, A_2, A_3\}$, $A_1 = \{p_1, p_2, p_5\}$, $A_2 = \{p_1, p_3, p_4\}$, $A_3 = \{p_1, p_4, p_5\}$ $d = 3$.

Step 2: $P' = \emptyset$.

Step 3: $S_1 = \{s_{1,1}, s_{1,2}, s_{1,3}\}$, $S_2 = \{s_{2,1}, s_{2,2}, s_{2,3}\}$, $S_3 = \{s_{3,1}, s_{3,2}, s_{3,3}\}$.

Table 1. The Shares of Six Participants by Using TUM Scheme

Participant p_i	The share of participant p_i
p_1	$s_{1,1}, s_{2,1}, s_{3,1}$
p_2	$s_{1,2}$
p_3	$s_{2,2}$
p_4	$s_{2,3}, s_{3,2}$
p_5	$s_{1,3}, s_{3,3}$
p_6	

In this case, shares are distributed as Table 1. The information rate of this example is $1/3$.

3 The proposed scheme

In this section, one perfect secret sharing scheme for r -uniform hypergraph-base access structure, called r -HA scheme, be proposed. Subsection 3.2 gives an example for easily understanding. Subsection 3.3 analysis the security and calculate the information rate of r -HA scheme.

3.1 r -HA Scheme

A one-way hash-function is a one-way function $h(x)$ which satisfy the following properties:

- (1) h maps bit-strings of arbitrary length to bit-strings of a fixed length.
- (2) When given h and an input x , $h(x)$ is easy to compute.
- (3) To find any pre-image x' such that $h(x') = y$ when given any y for which a corresponding input is not known.
- (4) It is computationally infeasible to find any second input which has the same output as any specified input.
- (5) It is computationally infeasible to find any two distinct inputs x, x' which hash to the same output.

Recall that in an r -uniform hypergraph-based access structure, the edge set is equal to Γ_0 . $\Gamma = \{A \subseteq P \mid S \subseteq A \text{ for any } S \in \Gamma_0\}$ and the prohibited structure is decided by $\Delta = 2^P \setminus \Gamma = \{A \subseteq P \mid S \not\subseteq A \text{ for all } S \in \Gamma_0\}$. Assume all computations are over Z_q where $q \geq \max\{K_0, K_1\}$ is a large prime for independent secrets K_0 and K_1 . The symbol $\oplus$ be used to express exclusive-or operation.

The distribution algorithm

Step 1: Let the secret key $K = \{K_0, K_1\}$ be taken from $[Z_q]^2$, and $k_2, \dots, k_{r-1}$ be randomly taken from $[Z_q]$.

Step 2: Construct a polynomial function: $f(x) = K_0 + K_1x + k_2x^2 + \dots + k_{r-1}x^{r-1}$.

Step 3: Compute $y_{(i_1, i_2, \dots, i_{r-1})} = f(\sum_{k=1}^{r-1} i_k n^{r-k-1})$ for $1 \leq$

$$i_1 < i_2 < \dots < i_{r-1} \leq n.$$

Step 4: Select n random numbers $a_1, a_2, \dots, a_n$ from Z_q .

Step 5: For $1 \leq i \leq n$, let $S_i = \langle a_{i,(1,2,\dots,r-1)}, \dots, a_{i,(j_1, j_2, \dots, j_{r-1})}, \dots, a_{i,(n-r+2, \dots, n)}, a_i \rangle$, for $1 \leq j_1 < j_2 < \dots < j_{r-1} \leq n$, where

$$a_{i,(j_1, \dots, j_{r-1})} = \begin{cases} y_{(j_1, \dots, j_{r-1})} + h(a_{j_1} \oplus a_{j_2} \oplus \dots \oplus a_{j_{r-1}}), & \text{if } \{p_i, p_{j_1}, \dots, p_{j_{r-1}}\} \in E(G); \\ \text{empty, otherwise.} \end{cases}$$

Step 6: Send S_i to participant p_i .

The reconstruction algorithm

Collect $S_{j_1}, S_{j_2}, \dots, S_{j_r}$ from $p_{j_1}, p_{j_2}, \dots, p_{j_r}$ together for $1 \leq j_1 < j_2 < \dots < j_r \leq n$, where $A = \{p_{j_1}, p_{j_2}, \dots, p_{j_r}\}$ is a minimal qualified subset. In the other words, $A \subseteq \Gamma_0$.

Step 1: For $1 \leq t \leq r$, let $\{i_1, i_2, \dots, i_{r-1}\} = \{j_1, j_2, \dots, j_r\} \setminus \{t\}$, compute $y_{(i_1, i_2, \dots, i_{r-1})} = a_{k, (i_1, i_2, \dots, i_{r-1})} - h(a_{i_1} \oplus a_{i_2} \oplus \dots \oplus a_{i_{r-1}})$.

Step 2: Collect r points $(\sum_{k=1}^{r-1} i_k n^{r-k-1}, y_{(i_1, i_2, \dots, i_{r-1})})$

from Step 1, $f(x) = K_0 + K_1x + k_2x^2 + \dots + k_{r-1}x^{r-1}$ can be reconstructed by using the Lagrange interpolating polynomial. And the secret key $K = \{K_0, K_1\}$ can be obtained.

3.2 An example for 3-HA scheme

This section gives an example for easily understanding the scheme that be stated in previous

subsection. There is a 3-uniform hypergraph $H = (V, E)$ in Figure 1 which denotes the minimal access structure with six participants. Where $V = \{p_1, p_2, p_3, p_4, p_5, p_6\} = P$, and $E = \{\{p_1, p_2, p_5\}, \{p_1, p_3, p_4\}, \{p_1, p_4, p_5\}\} = \Gamma_0$. That is, the access structure Γ and prohibited structure Δ are the same with the example that be stated in subsection 2.2.

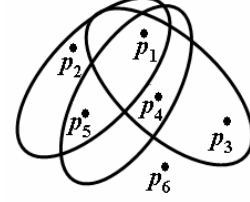


Figure 1. An example of 3-uniform hypergraph $H = (V, E)$.

Let the secret key $K = (K_0, K_1)$ is taken randomly from $Z_q \times Z_q$ and k_2 is taken randomly from Z_q , where q is a large prime. Let $f(x) = k_2x^2 + K_1x + K_0 \pmod{q}$. $y_{(i,j)}$ is computed from $f(x)$ as: $y_{(i,j)} = f(6i + j) \pmod{q}$, for $1 \leq i < j \leq 6$. A function h is a one way function and number $a_1, a_2, \dots, a_6$ are selected randomly over Z_q . The shares of participants are given by Table 2, where “-” denotes empty entry.

Table 2. The Shares of Six Participants by Using r -HA Scheme

share	The contain of share
S_1	$\langle -, -, -, -, -, -, y_{(2,5)} + h(a_2 \oplus a_5), -, y_{(3,4)} + h(a_3 \oplus a_4), -, -, y_{(4,5)} + h(a_4 \oplus a_5), -, -, a_1 \rangle$
S_2	$\langle -, -, -, y_{(1,5)} + h(a_1 \oplus a_5), -, -, -, -, -, -, -, -, -, a_2 \rangle$
S_3	$\langle -, -, y_{(1,4)} + h(a_1 \oplus a_4), -, -, -, -, -, -, -, -, -, -, a_3 \rangle$
S_4	$\langle -, y_{(1,3)} + h(a_1 \oplus a_3), -, y_{(1,5)} + h(a_1 \oplus a_5), -, -, -, -, -, -, -, -, -, a_4 \rangle$
S_5	$\langle y_{(1,2)} + h(a_1 \oplus a_2), -, y_{(1,4)} + h(a_1 \oplus a_4), -, -, -, -, -, -, -, -, -, -, a_5 \rangle$
S_6	$\langle -, -, -, -, -, -, -, -, -, -, -, -, -, -, a_6 \rangle$

In the following, we demonstrate the constructed secret sharing schemes satisfies:

- (1) If $A \subseteq P$ and $|A| < 3$, then A obtains no information regarding the secret;
- (2) If $e \notin A \subseteq P \forall e \in \Gamma_0$, and $|A| \geq 3$, A obtains no information regarding the secret;
- (3) If $\exists e \in \Gamma_0$, $e \subseteq A \subseteq P$ and $|A| \geq 3$, A can recover the secret.

For (1), given $A = \{p_1\}$, A can not obtain any information on all shares $y_{(i,j)}$. Therefore, A obtains no information about the secret K .

For (2), given $A = \{p_2, p_3, p_4, p_5, p_6\}$, since only random numbers a_2, a_3, a_4, a_6 are known, only $y_{(2,3)}, y_{(2,4)}, y_{(2,5)}, y_{(2,6)}, y_{(3,4)}, y_{(3,5)}, y_{(3,6)}, y_{(4,5)}, y_{(4,6)}$ and $y_{(5,6)}$ can be got. But A can not obtain any information on any one of them since S_2, S_3, S_4, S_5

and S_6 have no any information about these $y_{(i,j)}$. Therefore, A obtains no information about the secret K .

For (3), given $A = \{p_1, p_2, p_5\}$, A can recover the secret K because participants p_1 and p_2 can recover $y_{(1,2)}$, $y_{(1,5)}$ and $y_{(2,5)}$, and so the polynomial $f(x)$.

3.3 Security analysis and information rate for r -HA scheme

We prove three conditions for r -HA scheme:

- (1) If $A \subseteq P$ and $|A| < r$, then A obtains no information regarding the secret;
- (2) If $e \notin A \subseteq P$, for any $e \in \Gamma_0$, and $|A| \geq r$, A obtains no information regarding the secret;

(3) If there exists $e \in \Gamma_0$, $e \subseteq A \subseteq P$ and $|A| \geq r$, then A can recover the secret.

Theorem 1

For all $A \subseteq P$ and $|A| < r$, A obtain no information regarding the secret key of the r -HA scheme.

Proof. Without lose of generality, say $|A| = r - 1$ and assume $A = \{p_{j_1}, p_{j_2}, \dots, p_{j_{r-1}}\}$, where $1 \leq j_1 < j_2 < \dots < j_{r-1} \leq n$. The share of $S_i = \langle a_{i,(1,2,\dots,r-1)}, \dots, a_{i,(j_1, j_2, \dots, j_{r-1})}, \dots, a_{i,(n-r+2, \dots, n)}, a_i \rangle$ is the shares of p_i for $i \in \{j_1, j_2, \dots, j_{r-1}\}$. It is clear that no information about $y_{(j_1, j_2, \dots, j_{r-1})}$ for all $0 \leq j_1 < j_2 < \dots < j_{r-1} \leq n$, can be derived from $S_{j_1}, S_{j_2}, \dots, S_{j_{r-1}}$. Therefore, participants $p_{j_1}, p_{j_2}, \dots, p_{j_{r-1}}$ can not obtain any information about the secret key K . $\square$

Theorem 2

If $e \not\subseteq A \subseteq P$ for all $e \in \Gamma_0$ and $|A| \geq r$, A obtains no information regarding the secret key of the r -HA scheme.

Proof. Assume $A = \{p_{m_1}, p_{m_2}, \dots, p_{m_l}\}$, where $1 \leq m_1 < m_2 < \dots < m_l \leq n$ and $l \geq r$. Let $S_i = \langle a_{i,(1,2,\dots,r-1)}, \dots, a_{i,(j_1, j_2, \dots, j_{r-1})}, \dots, a_{i,(n-r+2, \dots, n)}, a_i \rangle$ is the shares of p_i for $i \in \{m_1, m_2, \dots, m_l\}$. Because $e \not\subseteq A \subseteq P$ for all $e \in \Gamma_0$, it is clear that no information about $y_{(j_1, j_2, \dots, j_{r-1})}$ for any $1 \leq j_1 < j_2 < \dots < j_{r-1} \leq n$ can be derived from $S_{m_1}, S_{m_2}, \dots, S_{m_l}$, no matter $\{j_1, j_2, \dots, j_{r-1}\} \subseteq \{m_1, m_2, \dots, m_l\}$ or not. Therefore, participants $p_{m_1}, p_{m_2}, \dots, p_{m_l}$ can not obtain any information of secret key K . $\square$

Theorem 3

If there exist $e \in \Gamma_0$ such that $e \subseteq A \subseteq P$ and $|A| \geq r$, A can recover the secret key of the r -HA scheme.

Proof. Assume $\{p_{m_1}, p_{m_2}, \dots, p_{m_r}\} = e \subseteq A$ for some $e \in \Gamma_0$, where $1 \leq m_1 < m_2 < \dots < m_r \leq n$. Let $S_i = \langle a_{i,(1,2,\dots,r-1)}, \dots, a_{i,(j_1, j_2, \dots, j_{r-1})}, \dots, a_{i,(n-r+2, \dots, n)}, a_i \rangle$ is the shares of p_i for $i \in \{m_1, m_2, \dots, m_r\}$. According to the construction of the distribution algorithm of r -HA scheme, it is clear that they can reconstruct $y_{(j_1, j_2, \dots, j_{r-1})}$ for $1 \leq j_1 < j_2 < \dots < j_{r-1} \leq n$ and $\{j_1, j_2, \dots, j_{r-1}\} \subseteq \{m_1, m_2, \dots, m_r\}$ (there are r such sub-secret $y_{(j_1, j_2, \dots, j_{r-1})}$ can be recover). Therefore participants in A can recover the secret key K . $\square$

Theorem 4

Suppose n is the number of vertices in an r -uniform hypergraph G and d is the maximum degree of G . The information rate ρ of the r -HA scheme based on G satisfies: $\rho = 2/(d + 1)$.

Proof. The share S_i for participant p_i is an $(C(n, r - 1) + 1)$ -dimensional vector, where $C(x, y) = x!/(y! \cdot (x - y)!)$. For the first $C(n, r - 1)$ dimensions, if $\{p_i, p_{l_1}, p_{l_2}, \dots, p_{l_r}\} \notin E(H)$ for $1 \leq l_1 < l_2 < \dots < l_r \leq n$,

$a_{i,(x_1, \dots, x_{r-1})}$ are empty for $\{i, x_1, x_2, \dots, x_{r-1}\} = \{l_1, l_2, \dots, l_r\}$. In other word, $a_{i,(x_1, \dots, x_{r-1})}$ s are over Z_q only when $\{p_i, p_{x_1}, p_{x_2}, \dots, p_{x_{r-1}}\} \in E(H)$ for $1 \leq x_1 < x_2 < \dots < x_{r-1} \leq n$. For the last one dimension, a_i always exists; Therefore, the share space S_i is equal to $[Z_q]^{\deg(p_i)+1}$, where $\deg(p_i)$ is the degree of vertex p_i in G . Hence, the maximal share space is $[Z_q]^{d+1}$, where d is the maximum degree of G . And because the secret space $\mathcal{K} = [Z_q]^2$, the information rate ρ of r -HA scheme based on G will satisfies $(2 \times \log q)/((d + 1) \times \log q) = 2/(d + 1)$. $\square$

4 Performance

This section compares the information rate between the proposed scheme and TUM scheme, and give some conclusions. At first, the information rate of TUM scheme in a special condition is calculated. Secondly, the information rate of the main scheme is greater than the information rate of TUM scheme for the same access structure will be proved.

Theorem 5

If $\max_{B \in \Delta} |B| \geq \max_{X \in \Gamma_0} |X|$, the information rate of TUM scheme is equal to $1/d$, where $d = \max_{p \in P} |\{X \in \Gamma_0 : p \in X\}|$.

Proof. Because $\max_{B \in \Delta} |B| \geq \max_{X \in \Gamma_0} |X|$, $P' = \emptyset$ at step 2 and $\Gamma_0' = \Gamma_0$ at step 1 in TUM scheme. Each participant p_i must get $|\{X \in \Gamma_0 : p_i \in X\}|$ shares at step 3 in TUM scheme. Hence, the maximal share space is $[Z_q]^d$. Note that the secret space $\mathcal{K} = Z_q$, the information rate ρ of TUM scheme will satisfies $(\log q)/((d \times \log q) = 1/d$. $\square$

Given an r -uniform hypergraph $G = (V, E)$. G is called *non-complete* if $|E| < C(n, r)$. The information rate of the proposed scheme is greater than the information rate of TUM scheme for any non-complete r -uniform hypergraph-based access structure. It is proved as following theorem.

Theorem 6

The information rate of the proposed scheme is greater than the information rate of TUM scheme for any non-complete r -uniform hypergraph-based access structure.

Proof. Since the given r -uniform hypergraph G is non-complete, $\max_{B \in \Delta} |B| \geq r = \max_{X \in \Gamma_0} |X|$. By Theorem 5, the information rate of TUM scheme, ρ_{TUM} , is $1/d$ where d is equal to the maximum degree in G . It is clearly that $d \geq 1$ and $d \times 2 = d + d \geq d + 1$ is true. Hence, according to Theorem 4, the information rate of the proposed scheme $= 2/(d + 1) > 1/d = \rho_{\text{TUM}}$ is always holds. $\square$

5 Conclusions

Given an r -uniform hypergraph G , this paper gives a construction of perfect secret sharing scheme for access structures based on G , called r -HA scheme. In G , a vertex denotes a participant, the edge set is equal to the minimal access structure Γ_0 and an edge is a set of r participants who can recover the secret. The information rate of r -HA scheme is equal to $1 / (d + 1)$, and it is shown by theorem 6 that the value is greater than $1 / d$ for any non-complete r -uniform hypergraph-based access structure, where d is the maximum degree of the r -uniform hypergraph G . In the other words, the proposed r -HA scheme is better than TUM scheme for any non-complete r -uniform hypergraph-based access structures. Note that for any complete r -uniform hypergraph-based access structure, $|\Gamma_0| = |E| = C(n, r)$, i.e., the access structure will become the access structure of an (r, n) -threshold scheme. In this case, Shamir's (r, n) -threshold scheme will be the best choice.

We can modify r -HA scheme by replace random numbers $k_2, k_3, \dots, k_{r-1}$ in this scheme by independent secret keys $K_2, K_3, \dots, K_{r-1}$. In this way, the secret key $K = (K_0, K_1, \dots, K_{r-1})$ and the information rate will be increasing to $r / (d + 1)$, but it will make this scheme not perfect anymore. One can choose the modified r -HA scheme or the original r -HA scheme according to their requirements.

Two secret sharing schemes for r -uniform hypergraph-based prohibited structure, called r -HP1 and r -HP2 scheme, were proposed by Weng and Juan in 2005 and 2006 [22, 23]. It is known that r -HP2 scheme is better than r -HP1 scheme. The proposed modified r -HA scheme and r -HP2 scheme are similar, but the access structures and prohibited structures they deal with are different mostly. Given an access structure Γ with $|S| = r$ for all $S \in \Gamma_0$. For all $S \subseteq P$ with $|S| = r + 1$, if there are $S' \in \Gamma_0$ such that $S' \subseteq S$, then the access structure will become $\Gamma = \Gamma_0 \cup \{A : A \subseteq P \text{ and } |A| \geq r + 1\}$. In this case, the specified access structure Γ can be solved by r -HP1, r -HP2, r -HA and modified r -HA schemes separately. Let $d = \max_{p \in P} |\{X \in \Gamma_0 : p \in X\}|$, the information rate of the modified r -HA scheme is $r/(d + 1)$, which is great than or equal to the information rate of r -HP2 scheme (since $r / (d + r + 1) \leq \rho(r\text{-HP2}) \leq r / (d + 1)$, see [23]). Hence, the modified r -HA scheme is better than r -HP2 scheme be concluded.

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